

CIDR & SUBNETTING

Need for Classless addressing

- The flaws in classful addressing scheme combined with the fast growth of the Internet led to the near depletion of the available addresses.
- Yet the number of devices on the Internet is much less than the 2^{32} address space.
- We have run out of class A and B addresses, and a class C block is too small for most midsize organizations.
- One solution that has alleviated the problem is the idea of classless addressing.

Need for classless addressing

Scenarios -

No. of requirements - 10 nodes, Address – Class C

Class C – 24 bits netid, 8 bits for host id

No of host id – $2^8 = 256$ nodes

Our request = 10 nodes = $256 - 10$ nodes = 246 id will be wasted

No.of Requirements – 2000 nodes, Address – Class A

Class A – 2^{24} nodes = 2000 nodes of – **16777216-2048**
node ids will be wasted

CLASSLESS ADDRESSING

- ★ Formal name is Classless Inter-Domain Routing (CIDR).
- ★ Created a new set of standards that allowed service providers to allocate IPv4 addresses on any address bit boundary (prefix length) instead of only by a class A, B, or C address.
- ★ Classless addressing is possible with the help of subnetting.

IPv4 –CIDR NOTATION

- CIDR Block

When a user asks for specific number of IP Addresses,

- CIDR dynamically assigns a block of IP Addresses based on certain rules.
- This block contains the required number of IP Addresses as demanded by the user.
- This block of IP Addresses is called as a CIDR block.

IPv4 –CIDR NOTATION

- Classless Addressing-
- Classless Addressing is an improved IP Addressing system.
- It makes the allocation of IP Addresses more efficient.
- It replaces the older classful addressing system based on classes.
- It is also known as Classless Inter Domain Routing (CIDR).

To remember

- No of Blocks --- No .of Network
- Block size --- Host size --- No.of addresses
- 2 types
- VLSM – Variable Length Subnet mask
- FLSM– Fixed Length Subnet mask – Class addressing

Rules For Creating CIDR Block

- A CIDR block is created based on the following 2 rules

Rule-01:

- All the IP Addresses in the CIDR block must be contiguous.

Rule-02:

- The size of the block must be presentable as power of 2.
- Size of the block is the total number of IP Addresses contained in the block.
- Size of any CIDR block will always be in the form 2ⁿ
- 1
- 2²
- 2³
- 2⁴
- 2⁵

and so on

IPv4 –CIDR NOTATION

- In IPV4 addressing a block of addresses can be defined as x.y.z.w/n in which x.y.z.w defines one of the addresses and /n defines the mask
- The first address in the block can be found by setting the rightmost 32-n bits to 0s
- The last address in the block can be found by setting the rightmost 32-n bits to 1s
- So the number possible addresses $= 2^{32-n}$

1. A block of addresses is granted to a small organization. One of the addresses is 205.16.37.39/28.
 - a. What is the first address in the block?
 - b. What is the last address in the block?
 - c. What subnet mask would be used by a station within the organization?

*A block of addresses is granted to a small organization.
We know that one of the addresses is 205.16.37.39/28.
What is the first address in the block?*

Solution

The binary representation of the given address is

11001101 00010000 00100101 00100111

If we set 32–28 rightmost bits to 0, we get

11001101 00010000 00100101 00100000

or

205.16.37.32.

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If we set 32–28 rightmost bits to 0, we get

11001101 00010000 00100101 00100000

or

205.16.37.32.

- Subnet mask
- 11111111 11111111 11111111 11110000
- AND
- 11001101 00010000 00100101 00100111

11001101 00010000 00100101 00100000

Or

205.16.37.32

Find the last address for the block in Example 19.6.

Solution

The binary representation of the given address is

11001101 00010000 00100101 00100111

If we set 32 – 28 rightmost bits to 1, we get

11001101 00010000 00100101 00101111

or

205.16.37.47

Find the number of addresses in Example 19.6.

Solution

The value of n is 28, which means that number of addresses is 2^{32-28} or 16.

Example 11

A small organization is given a block with the beginning address and the prefix length 205.16.37.24/29 (in slash notation). What is the range of the block?

Solution

The beginning address is 205.16.37.24. To find the last address we keep the first 29 bits and change the last 3 bits to 1s.

Beginning: 11001111 00010000 00100101 00011000

Ending : 11001111 00010000 00100101 00011111

There are only 8 addresses in this block.

Example 5.29

One of the addresses in a block is 110.23.120.14/20. Find the number of addresses, the first address, and the last address in the block.

Solution

The network mask is 255.255.240.0.

- a. The number of addresses in the network is $2^{32-20} = 4096$.
- b. To find the first address, we apply the first short cut to bytes 1, 2, and 4 and the second short cut to byte 3. The first address is 110.23.112.0/20.

Address:	110	.	23	.	120	.	14
Network mask:	255	.	255	.	240	.	0
First address (AND):	110	.	23	.	112	.	0

For the IP address 192.168.200.0/24

Find the number of hosts, first address and last address

Total IPv4 Address Bits = 32

Network Prefix (Network Bits) = 24

Host Bits (h) = $32 - 24 = 8$

1. **Total IP addresses with /24 prefix?**

Total Host Address formula = $2^h = xx$

Putting the value = $2^8 = 256$

So, the address range start from 192.168.200.0 – 192.168.200.255

2. **Total usable IP addresses with /24 prefix?**

Usable Host Address Formula = $2^8 - 2 = xx$

Putting the value = $2^8 - 2 = 254$

3. **The first address are ever network address in this example
192.168.200.0 is network address.**

4. **The last address are ever the broadcast address, in this
example the address 192.168.200.255 is broadcast
address.**

- What is the broadcast address for this block of addresses?
- What is the network address for this block of addresses?
- What is the first host address in this block (in dotted decimal representation)?
- What is the last host address in this block (in dotted decimal representation)?

192.32.190.18/21

Network ID	host ID
1100 0000.0010 0000.1011 1	1 10.0001 0010

a) 1.5pts

```
broadcast Address:    1100 0000. 0010 0000. 1011 1 111. 1111 1111
                        192 . 32 . 191 . 255
```

b) 1.5pts

Network Address: 1100 0000.0010 0000.1011 1 000. 0000 0000
192 . 32 . 184 . 0

c) 1.5pts

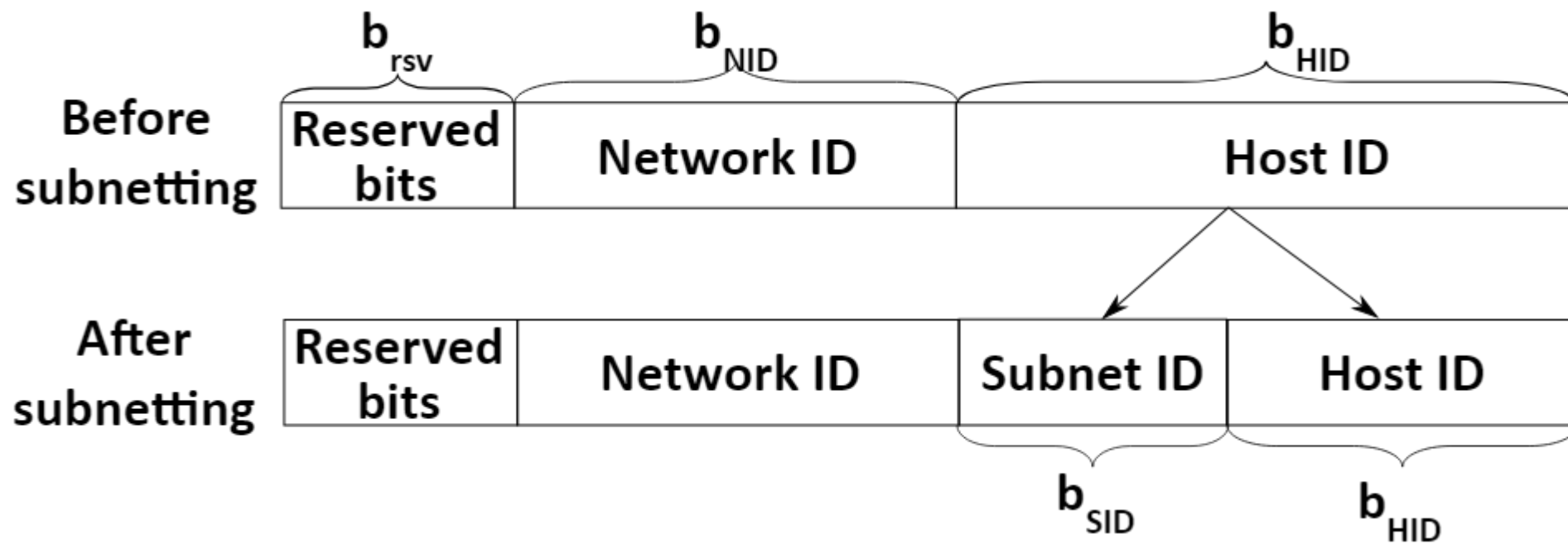
First host Address: 1100 0000.0010 0000.1011 1 000. 0000 0001
192 . 32 . 184 . 1

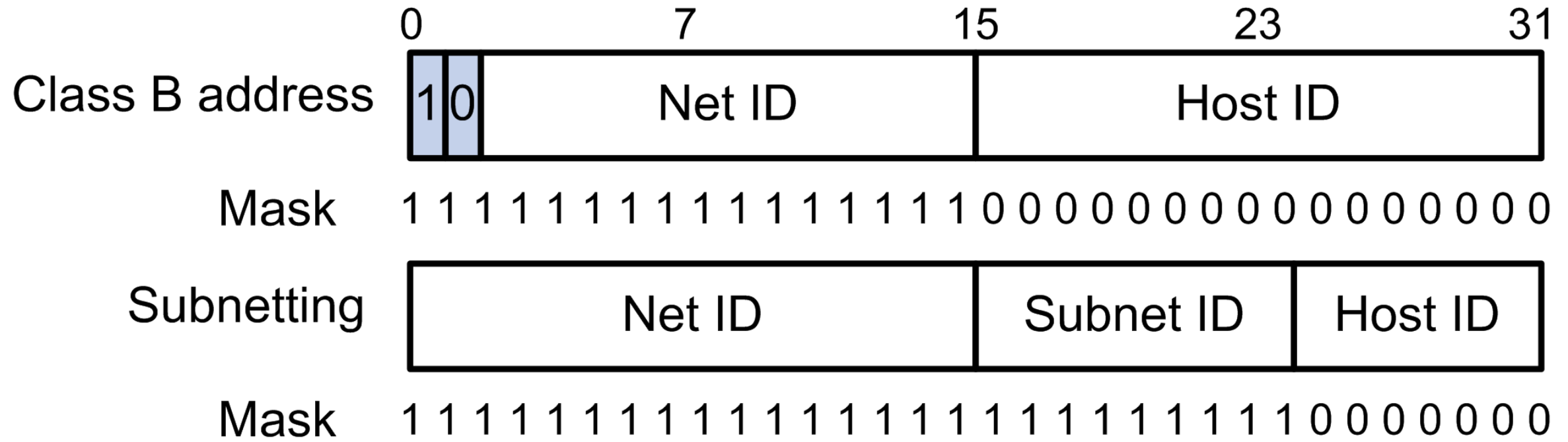
d) 1.5pts

Last host Address: 1100 0000. 0010 0000. 1011 1 111. 1111 1110
192 . 32 . 191 . 254

Find the Number of Subnets & the Number of Valid Hosts per Subnet

- Subnets
 - $2^{\text{\# of subnet bits}}$
- Valid Hosts
 - $2^{\text{\# of host bits}} - 2$





Subnet Mask 255.255.255.192

	26 bits for Network ID			6 bits for Host ID
Decimal	255	255	255	192
Binary	11111111	11111111	11111111	11000000

192.168.1.0



255.255.255.240 (/28)

11111111.11111111.11111111.11110000

192.168.1.0

255.255.255.240 (/28)

11111111.11111111.11111111.11110000

Network

- Is this a Class A, B, or C network?

- Subnets = $2^4 = (2 \times 2 \times 2 \times 2) = 16$

- Hosts = $2^4 = 16 - 2 = 14$

150.150.0.0

 255.255.255.252 (/30)
11111111.11111111.11111111.11111100

150.150.0.0

255.255.255.252 (/30)

11111111.11111111.11111111.11111100

Network

- Is this a Class A, B, or C network?

- Subnets = $2^{14} = 16,384$

- Hosts = $2^2 = 4 - 2 = 2$

Example

- For the ip address 192.168.2.0/27 in class addressing and the subnet mask is 255.255.255.224

Find the total number of hosts in each network

Find the total number of subnets

List the subnet addresses

To remember : class c and subnet mask is 255.255.255.0

Example

- For the ip address 192.168.2.0/27 in class addressing and the subnet mask is 255.255.255.224

Find the total number of hosts in each network

Find the total number of subnets

List the subnet addresses

$$\begin{aligned}\text{Total number of hosts} &= 32 - 27 = 5 \\ &= 2^5 \\ &= 32 \text{ hosts}\end{aligned}$$

Subnets??

224 = 11100000 = LSB 5 bits are for the host and 111-- 3 bits denote the subnet

- For the ip address 192.168.2.0/27 in class addressing and the subnet mask is 255.255.255.224

Find the total number of subnets

List the subnet addresses

$$\begin{aligned}\text{Total number of hosts} &= 32 - 27 = 5 \\ &= 2^5 \\ &= 32 \text{ hosts}\end{aligned}$$

Subnets??

224 = 11100000 = LSB 5 bits are for the host and 111 3 bits denote the subnet

total number of subnets = $2^3 = 8$ subnets with 32 hosts

- Subnet Mask: 255.255.255.224

- Formula: $256 - (\text{octet} \neq 255 \text{ or } 0)$

32

- 192.168.2.0/27

- 192.168.2.32/27

- 192.168.2.64/27

- 192.168.2.96/27

- 192.168.2.128/27

- 192.168.2.160/27

- 192.168.2.192/27

- 192.168.2.224/27

+32

8

Example 2

- For the ip address 172.16.0.0/27 in class addressing
Find the total number of hosts in each network
Find the total number of subnets
List the subnet addresses

To remember : class B and subnet mask is 255.255.0.0

Example 2

- For the ip address 172.16.0.0/27 in class addressing

Find the total number of hosts in each network

Find the total number of subnets

List the subnet addresses

$$\begin{aligned}\text{Total number of hosts} &= 32-27 = 5 \\ &= 2^5 \\ &= 32 \text{ hosts}\end{aligned}$$

Subnets??

Example 2

- For the ip address 172.16.0.0/27 in class addressing
Find the total number of hosts in each network
Find the total number of subnets
List the subnet addresses

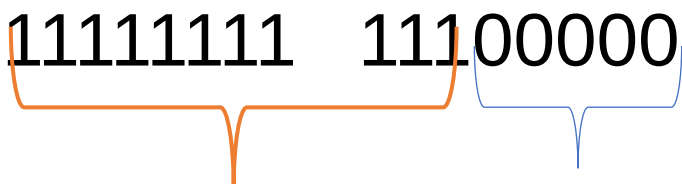
Subnets??

Class B – 255.255.0.0

Network id is 172.16

To find subnets as per 27 MSB are 1

11111111 11111111 11111111 11100000



subnet

host id bits

- For the ip address 172.16.0.0/27 in class addressing

Find the total number of hosts in each network

Find the total number of subnets

List the subnet addresses

$$\begin{aligned}\text{Total number of hosts} &= 32-27 = 5 \\ &= 2^5 \\ &= 32 \text{ hosts}\end{aligned}$$

$$\text{Subnets??} = 2^{11} =$$

2048 subnets

172.16.0.0 255.255.0.0

- Subnet Mask: 255.255.255.224
 • Formula: $256 - 224 = 32$ (11)
 • 172.16.0.0/27 ←
 • 172.16.0.32
 • 172.16.0.64
 • 172.16.0.96
 • 172.16.0.128
 • 172.16.0.160
 • 172.16.0.192
 • 172.16.0.224
- 172.16.1.0/27
 • 172.16.1.32/27 —
 • 172.16.1.64/27 —
 • 172.16.1.96/27 —
 • 172.16.1.128/27
 • 172.16.1.160/27
 • 172.16.1.192/27
 • 172.16.1.224/27
- 172.16.2.0/27
 ↑
 255
- 11 subnet bits
 $2^{11} = 2,048$ subnets

• Subnet Mask: 255.255.255.224

• Formula: $256 - \underline{224} = \underline{32}$

172.16.2.0/27

25'

• 172.16.0.0/27 ←

• 172.16.0.32/27

• 172.16.0.64/27

• 172.16.0.96/27

• 172.16.0.128/27

• 172.16.0.160/27

• 172.16.0.192/27

• 172.16.0.224/27

• 172.16.1.0/27

• 172.16.1.32/27 —

• 172.16.1.64/27 —

• 172.16.1.96/27 —

• 172.16.1.128/27

• 172.16.1.160/27

• 172.16.1.192/27

• 172.16.1.224/27

172.16.0.0 255.255.0.0

- Subnet Mask: 255.255.255.224

172.16.2.0/27

255

- Formula: 2^n

Last subnet:

172.16.255.224/27

- 172.16.0.0/27
- 172.16.0.32/27 —
- 172.16.0.64/27 —
- 172.16.0.96/27 —
- 172.16.0.128/27
- 172.16.0.160/27
- 172.16.0.192/27
- 172.16.0.224/27
- 172.16.1.0/27
- 172.16.1.32/27
- 172.16.1.64/27
- 172.16.1.96/27
- 172.16.1.128/27
- 172.16.1.160/27
- 172.16.1.192/27
- 172.16.1.224/27

Valid Subnet Masks

[illegible]

<i>/n</i>	<i>Mask</i>	<i>/n</i>	<i>Mask</i>	<i>/n</i>	<i>Mask</i>	<i>/n</i>	<i>Mask</i>
/1	128.0.0.0	/9	255.128.0.0	/17	255.255.128.0	/25	255.255.255.128
/2	192.0.0.0	/10	255.192.0.0	/18	255.255.192.0	/26	255.255.255.192
/3	224.0.0.0	/11	255.224.0.0	/19	255.255.224.0	/27	255.255.255.224
/4	240.0.0.0	/12	255.240.0.0	/20	255.255.240.0	/28	255.255.255.240
/5	248.0.0.0	/13	255.248.0.0	/21	255.255.248.0	/29	255.255.255.248
/6	252.0.0.0	/14	255.252.0.0	/22	255.255.252.0	/30	255.255.255.252
/7	254.0.0.0	/15	255.254.0.0	/23	255.255.254.0	/31	255.255.255.254
/8	255.0.0.0	/16	255.255.0.0	/24	255.255.255.0	/32	255.255.255.255

Subnet Mask (Decimal)	Subnet Mask (Binary)	Valid/ Invalid
255.255.255.240		
255.230.255.0		
255.255.0.0		
240.0.0.0		
223.0.0.0		
255.0.255.0		

Valid and Invalid Subnet Masks

Subnet Mask (Decimal)	Subnet Mask (Binary)	Valid/ Invalid
255.255.255.240	1111111.1111111.1111111.11110000	Valid
255.230.255.0	1111111.11100110.1111111.00000000	Invalid
255.255.0.0	1111111.1111111.00000000.00000000	Valid
240.0.0.0	11110000.00000000.00000000.00000000	Valid
223.0.0.0	1101111.00000000.00000000.00000000	Invalid
255.0.255.0	1111111.00000000.1111111.00000000	Invalid

- Subnet will be consecutive 1s only and not mixture of 1s and 0s

Block Allocation in classless addressing

- How are the blocks allocated?
- responsibility of block allocation is given to a global authority called the Internet Corporation for Assigned Names and Addresses (ICANN).
- ICANN does not normally allocate addresses to individual Internet users.
- It assigns a large block of addresses to an ISP.
- For the proper operation of the CIDR, three restrictions need to be applied to the allocated block.

Rules for classless

1. Number of hosts/block size/no.of addresses = 2^n
2. Number of subnets= 2^n
3. Subnet will be consecutive 1s only and not mixture of 1s and 0s
4. However, there is a restriction on choosing the beginning addresses of the block.
 - beginning address needs to be divisible by the number of addresses in the block.
 - To see this restriction, we can show that the beginning address can be calculated as $(X \times 2^{32-n})$ in which X is the decimal value of the prefix. In other words, the beginning address is $X \times N$.

Fix CIDR – Examples

- 10.10.1.0
- How many address – $4 = 2^2$
- 10.10.1.0, 10.10.1.1, 10.10.1.2, 10.10.1.3
- Size of Host id = 2 bits
- Size of network id is = $32 - 2 = 30$ bits
- 10.10.1.0

- How many address – 100 nodes = $2^7 = 128$ address
- 10.10.1.0 /25 to 10.10.1.127/25
- Size of the host id = 7
- Size of the net id = $32 - 7 = 25$ – CIDR value
- 10.10.1.5/25

Fix CIDR – Examples

- XYZ = 32 hosts – 5bits
- 10.10.1.0 to 10.10.1.31
- Host id = 5 bits
- Netid = $32 - 5 = 27$
- CIDR = 27
- 10.10.1.0/27

Subnetting – 32 address

- 10.10.1.5/27
- AND operation on
- 00001010.00001010.00000001.00000101
- 11111111.11111111.11111111.11100000
- -----
- 00001010.00001010.00000001.00000000
- 255.255.255.224 – subnet mask
- Why do we need subnet masking?

Example

An ISP is granted a block of addresses starting with 190.100.0.0/16. The ISP needs to distribute these addresses to three groups of customers as follows:

1. The first group has 64 customers; each needs 256 addresses.
2. The second group has 128 customers; each needs 128 addresses.
3. The third group has 128 customers; each needs 64 addresses.

Design the subblocks and give the slash notation for each subblock. Find out how many addresses are still available after these allocations.

Solution

Group 1

For this group of 64 customers, each customer needs 256 addresses. This means the suffix length is 8 ($2^8 = 256$). The prefix length is then $32 - 8 = 24$.

01: 190.100.0.0/24 → 190.100.0.255/24

02: 190.100.1.0/24 → 190.100.1.255/24

.....

64: 190.100.63.0/24 → 190.100.63.255/24

Total = $64 \times 256 = 16,384$

Solution (Continued)

Group 2

For this group of 128 customers, each customer needs 128 addresses. This means the suffix length is 7 ($2^7 = 128$). The prefix length is then $32 - 7 = 25$. The addresses are:

001: 190.100.64.0/25 → 190.100.64.127/25

002: 190.100.64.128/25 → 190.100.64.255/25

.....

127: 190.100.127.0/25 → 190.100.127.127/25

128: 190.100.127.128/25 → 190.100.127.255/25

Total = $128 \times 128 = 16,384$

Solution (Continued)

Group 3

For this group of **128 customers**, each customer needs 64 addresses. This means the suffix length is 6 ($2^6 = 64$). The prefix length is then $32 - 6 = 26$.

001:190.100.128.0/26 ➔ 190.100.128.63/26

002:190.100.128.64/26 ➔ 190.100.128.127/26

.....

128:190.100.159.192/26 ➔ 190.100.159.255/26

Total = $128 \times 64 = 8,192$

Solution (Continued)

Number of granted addresses: 65,536

Number of allocated addresses: 40,960

Number of available addresses: 24,576

The available addresses range from:

190.100.160.0 **→** 190.100.255.255

Total = $96 \times 256 = 24,576$

Given the IP address 19.0.0.0. Design the 1000 subnets and give the range of 345th subnet and 632th subnet

- 19.0.0.0- Class A- Default Subnet mask: 255.0.0.0
- Netid : 8 bits hostid: 24 bits
- No of subnets needed is $1000=1024=2^{10}$
- Borrow 10 bits for subnetid from hostid : Netid:8 bits
- Subnetid: 10 bits and Hostid: 14 bits
- Make all netid and subnetidbits to 1
- NNNNNNNN SSSSSSSS SSHHHHHH HHHHHHHH
- 11111111 11111111 11000000 00000000
- 255.255.192.0- New subnet mask

19.0.0.0/18- it can be represented as

1st subnet (1-1 =0)

NNNNNNNN SSSSSSSS SSHHHHHH
HHHHHHHH

First address

19.00000000 00000000 00000000

19.0.0.0/18

Last address

19.00000000 . 00111111 11111111

19.0.63.255/18

.
. .
.

Last subnet – 1024 (1024-1=1023)

First address

19.11111111.11000000 00000000

19.255.192.0/18

Last address

19.11111111.11111111 111111 11111111

19.255.255.255/18

give the range of 345th subnet and 632th subnet

345th subnet (345-1=344)

Binary value of 344 =101011000

First address

19.01010110 . 00000000 000000000

19.86.0.0/18

Last address

19. 01010110 . 00111111 11111111

19.86.63.255/18

.

632(632-1=631)

Binary value of 631=1001110111

First address

19.10011101. 11000000 00000000

19.157.192.0/18

Last address

10011101 11111111 11111111

19.157.255.255/18